IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 17 July 2025, accepted 17 August 2025, date of publication 21 August 2025, date of current version 28 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3601349

RESEARCH ARTICLE

A Digital Twin Approach to Smart Monitoring and Fault Diagnosis

ANA C. S. ARAÚJO[1], LANE M. RABELO[2], (Member, IEEE), PAULO C. M. LAMIM FILHO[2], AND WALMIR M. CAMINHAS[3]

[1]Graduate Program in Electrical Engineering, Federal University of Minas Gerais, Belo Horizonte, Minas Gerais 31270-901, Brazil
[2]Department of Electrical Engineering/DEPEL, Federal University of São João del Rei, São João del Rei, Minas Gerais 36307-352, Brazil
[3]Department of Electronics Engineering, Federal University of Minas Gerais, Belo Horizonte, Minas Gerais 31270-901, Brazil

Corresponding author: Lane M. Rabelo (rabelo@ufsj.edu.br)

This work was supported in part by the State of Minas Gerais Research Foundation (FAPEMIG), the National Research Council (CNPq), Coordination for the Improvement of Higher Education Personnel and Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES), and the National Institute of Science and Technology in Electric Energy (INERGE); and in part by the Article Processing Charge was funded by CAPES under Grant ROR identifier: 00x0ma614.

ABSTRACT The continuous monitoring of induction motors, combined with early fault detection, is an essential tool for reducing maintenance costs and preventing unexpected downtime. This paper presents a Digital Twin-based methodology for the real-time diagnosis of incipient inter-turn short-circuit faults in induction motors. This diagnostic strategy capitalizes on the digital twin's real-time integration of a physical asset and its virtual counterpart. It enables the continuous comparison of measured and simulated currents through residual analysis. The Evolving Similarity-Based Modeling Plus (eSBM+) approach facilitates the identification of the motor's current operating state by comparing it with previously recorded conditions, supporting the precise detection of the onset and evolution of anomalies. The Reconstruction-Based Contributions Online (RBC-Online) system provides real-time insights into the root cause of the fault. The deviation identification automatically triggers a notification to the responsible technician, ensuring timely and well-informed decision-making. Experimental results validated the effectiveness of the proposed methodology, demonstrating high sensitivity and robustness under varying load conditions.

INDEX TERMS Digital twin, fault diagnosis, induction motors, inter-turn short circuit, real-time monitoring.

I. INTRODUCTION

The concept of Digital Twin (DT) has gained prominence in the industry as an essential technology for digital transformation [1], [2], [3]. This tool entails the creation of a highly detailed virtual replica of physical assets, systems, or processes, facilitating continuous monitoring and the analysis of various operational scenarios [4], [5].

A major benefit of DT lies in its ability to bridge the gap between the physical and digital worlds, enabling real-time data acquisition and advanced analytics [5], [6]. Through the integration of sensor data, historical records, and predictive models, DT delivers valuable insights into equipment performance, efficiency, and potential failure points [3]. This proactive approach enhances operational decision-making, reduces unplanned downtime, and optimizes maintenance strategies by shifting from corrective to predictive maintenance [7].

The convergence of Artificial Intelligence (AI) and the DT paradigm has established a robust and forward-looking approach for predicting and analyzing degradation patterns in electric motors, playing a pivotal role in fault diagnosis [8], [9], [10].

AI-based methods have gained prominence in industrial maintenance, as they enable the replication of human reasoning in decision-making and the execution of complex tasks [11]. These techniques encompass a wide range of approaches aimed at fault detection and classification, which can be categorized into methods based on historical data and approaches independent of prior data [8], [12], [13].

In data-driven methods, a labeled dataset represents a foundational requirement. This dataset must encompass

The associate editor coordinating the review of this manuscript and approving it for publication was Padmanabh Thakur.

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comprehensive process data, operational state labels, and the specific parameters associated with the modeling approach [14]. Furthermore, these techniques demand a dedicated training phase using extensive and representative data to achieve reliable predictive accuracy and diagnostic performance. Nevertheless, obtaining real-world failure data from electric motors remains a significant obstacle, as it typically requires the equipment to operate under fault conditions. This scenario is impractical and undesirable in most industrial contexts [15], [16].

Given these limitations, evolutionary approaches have gained considerable attention in recent decades due to their capacity for incremental learning and real-time adaptation [17]. These methods construct a dynamic knowledge base by continuously monitoring operational data and organizing information that characterizes various states of motor operation conditions. As a result, the system can iteratively refine its internal representations, enabling continuous monitoring and automated fault detection. It permits rapid identification and communication of potential anomalies to maintenance personnel, allowing quick and effective corrective actions [17], [18].

Evolutionary methods can be implemented in both supervised and unsupervised frameworks, offering enhanced flexibility and efficiency for the predictive monitoring of industrial equipment [15], [19], [20].

Unsupervised learning-based approaches are particularly advantageous, as labeling information is not always available. The Evolving Similarity-Based Modeling Plus (eSBM+) method relies on a model construction that incrementally learns from observed data, grouping similar information to represent system behavior under different operational conditions [12]. One of the main advantages of eSBM+ is its ability to operate online, allowing the model to continuously update itself without requiring prior training. Additionally, unlike conventional approaches that rely on large labeled datasets, eSBM+ can adapt to scenarios where historical data availability is limited or nonexistent. This characteristic makes the method particularly suitable for predictive maintenance applications and anomaly detection in electric motors.

Beyond fault detection and classification tasks, supervisors must understand the causes that led to a defect. In cases of induction motor failure, determining the root cause can be a complex and time-consuming process, making problem resolution difficult and potentially compromising both system performance and operational safety [21]. Delays in this diagnosis can negatively impact operational continuity, reduce system efficiency, and compromise process safety [22].

In response to this challenge, the Reconstruction-Based Contributions Online (RBC-Online) method assists in the rapid and accurate identification of fault causes by analyzing the contributions of each monitored variable in forming a new cluster. This tool provides a significant advantage in optimizing maintenance processes and mitigating industrial risks, improving the reliability and efficiency of electric motor diagnostics [12].

To further improve fault diagnosis in industrial applications, integrating these advanced techniques offers significant benefits for induction motors, which are widely utilized in various sectors due to their robustness and efficiency [23], [24], [25].

However, induction motors are susceptible to different types of failures, with short circuits in the stator winding contributing to approximately 28% to 36% of induction motor failures [26]. Among electrical faults, inter-turn short circuits are especially critical [27], [28], [29].

By applying AI-driven models like eSBM+ and fault attribution techniques such as RBC-Online, it becomes possible to accurately monitor the operational state of induction motors, detect early signs of anomalies, and implement predictive maintenance strategies. This proactive approach reduces unplanned downtime and enhances equipment lifespan, ensuring increased reliability in industrial operations [30].

In this context, this study develops a Digital Twin for induction motors to simulate precisely their healthy operating conditions under varying load torque. Continuous comparison of real-time signals from the physical and virtual systems enables the worker to identify the early stages of inter-turn short circuits. The system employs the eSBM+ method to recognize patterns consistent with previously observed anomalies, isolate the time frame in which the event occurs, and, through the RBC-Online method, actively determine the root cause of the fault.

II. METHODOLOGY

The methodology is illustrated in Fig. 1. Communication among the system's components is managed via the Message Queuing Telemetry Transport (MQTT) protocol, facilitating efficient and lightweight message exchange. The process initiates with the data processor transmitting a control message to trigger data acquisition.

The sensors connected to the ESP32 microcontroller capture the measurements and segment the data into small packets to optimize transmission efficiency. These packets are sent over a Wi-Fi connection to the processing unit, which receives, organizes, and executes the necessary computations to analyze the collected information.

Data processing takes place in the cloud. Upon processing, the results are forwarded to the data server and stored in the cloud using the InfluxDB time-series database. Finally, the stored data is accessible to the maintenance team through the Grafana interface, enabling remote and local monitoring of the induction motor's operational status.

MQTT and InfluxDB form a solid foundation for meeting the performance, flexibility, and scalability requirements of the proposed system. MQTT implements a lightweight and efficient publish/subscribe communication model that enables reliable, low-latency data transmission across bandwidth-constrained networks. Its architecture makes it

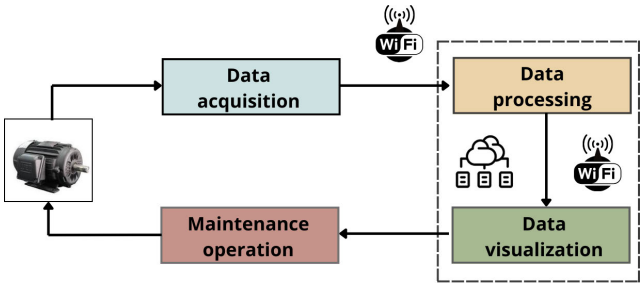


FIGURE 1. Flowchart of the methodology.

especially effective in industrial environments characterized by distributed sensor networks that continuously generate high-frequency data streams [31].

InfluxDB facilitates high-throughput time-series ingestion and efficient querying [32], essential for real-time monitoring and analytics in Digital Twin systems. Both technologies support horizontally scalable architectures with minimal configuration effort, allowing the system to manage increasing data volumes and analytical complexity without compromising processing speed or data integrity.

Fig. 2 presents the expansion of the proposed methodology for monitoring the operational condition of the induction motor, aiming at the diagnosis of inter-turn short circuits. Each block will be further detailed in the following subsections.

A. PHYSICAL ENTITY

This study analyzes a physical system consisting of a three-phase induction motor rated at 1.5 kW, operating at 220 V and 60 Hz, with four poles and a nominal speed of 1750 rpm. The motor drives a mechanically coupled DC generator, which supplies power to a variable resistance bank that serves as the load. Fig. 3 shows the test bench.

For signal acquisition, the authors designed and built a dedicated prototype capable of capturing voltage and current waveforms. An ESP32 microcontroller functions as the core processing unit. Upon receiving a control message containing its identifier—sent by the local machine via the MQTT protocol—the ESP32 begins acquiring the signals. The system uses an architecture that includes an intermediate storage mechanism. In this setup, the microcontroller temporarily stores the acquired data in local memory before transmitting it in structured blocks (packets) to a remote destination. This segmentation and local storage strategy ensures better control over data integrity. In cases of partial communication loss, the ESP32 retains the data, enabling retransmission and enhancing the reliability of the acquisition and transmission process.

For voltage and current signals, the system organizes the data into 100 packets, with each packet containing 20 samples from each corresponding vector. The ESP32's analog-to-digital converter (ADC) is susceptible to electrical noise, which may result in significant variations in its readings. To reduce this effect, the use of a bypass capacitor—typically a 100 nF ceramic capacitor—connected directly to the ADC input pin is recommended, depending on the application scenario [33]. In this work, a capacitor was incorporated into the acquisition circuit to enhance signal stability and mitigate noise-related distortions.

TABLE 1. Motor parameters.

$R_s(\Omega)$	$R_r(\Omega)$	$X_s(\Omega)$	$X_r(\Omega)$	$X_m(\Omega)$
5.40	2.030	7.367	4.273	137.826

The system employs SCT-013 and ZMPT101B sensors for current and voltage measurements, respectively. The SCT-013-20A sensor measures currents up to 20 A (RMS), with an analog output of 1 V (RMS). The ZMPT101B sensor incorporates an internal potential transformer, allowing voltage measurements up to 250 V (RMS) and providing a proportional analog signal to the microcontroller.

B. VIRTUAL ENTITY

Researchers have developed induction motor models to address different analytical and application requirements. One of the most traditional and widely used approaches is the positive-sequence equivalent circuit, typically formulated in the dq reference frame fixed to the stator. Its structure is illustrated in Fig. 4. The stator resistance and leakage reactance are denoted by R_s and X_s, respectively. The magnetizing branch is characterized by the magnetization reactance X_m. On the rotor side (referred to as the stator), the resistance and leakage reactance are represented by R_r and X_r.

This study defines the virtual entity based on the per-phase steady-state equivalent circuit of the induction motor. The authors obtained the motor's electrical parameters—including resistances and inductances—by performing standard tests: the no-load test (open-circuit) and the short-circuit (blocked-rotor) test. Table **??** presents the parameters extracted from these experimental procedures.

Determining the machine's slip during induction motor monitoring requires knowledge of its rotational speed. This study employs a speed estimator that derives the motor speed from the voltage and current signals measured from the physical system.

The rotor speed (ω_r) is typically determined by subtracting the slip frequency (ω_s) from the synchronous frequency (ω). The slip frequency, in turn, is derived from the stator flux (ϕ) and the stator currents, as expressed in (1). In this equation, R_r represents the rotor resistance, while L_m, L_s, and L_r correspond to the mutual, stator, and rotor inductances, respectively. The subscripts ds and qs refer to the stator components in the d-q reference frame.

$$\omega_s = \frac{R_r i_{qs}}{\phi_s - \left(L_s - \frac{L_m^2}{L_r}\right) i_{ds}} \tag{1}$$

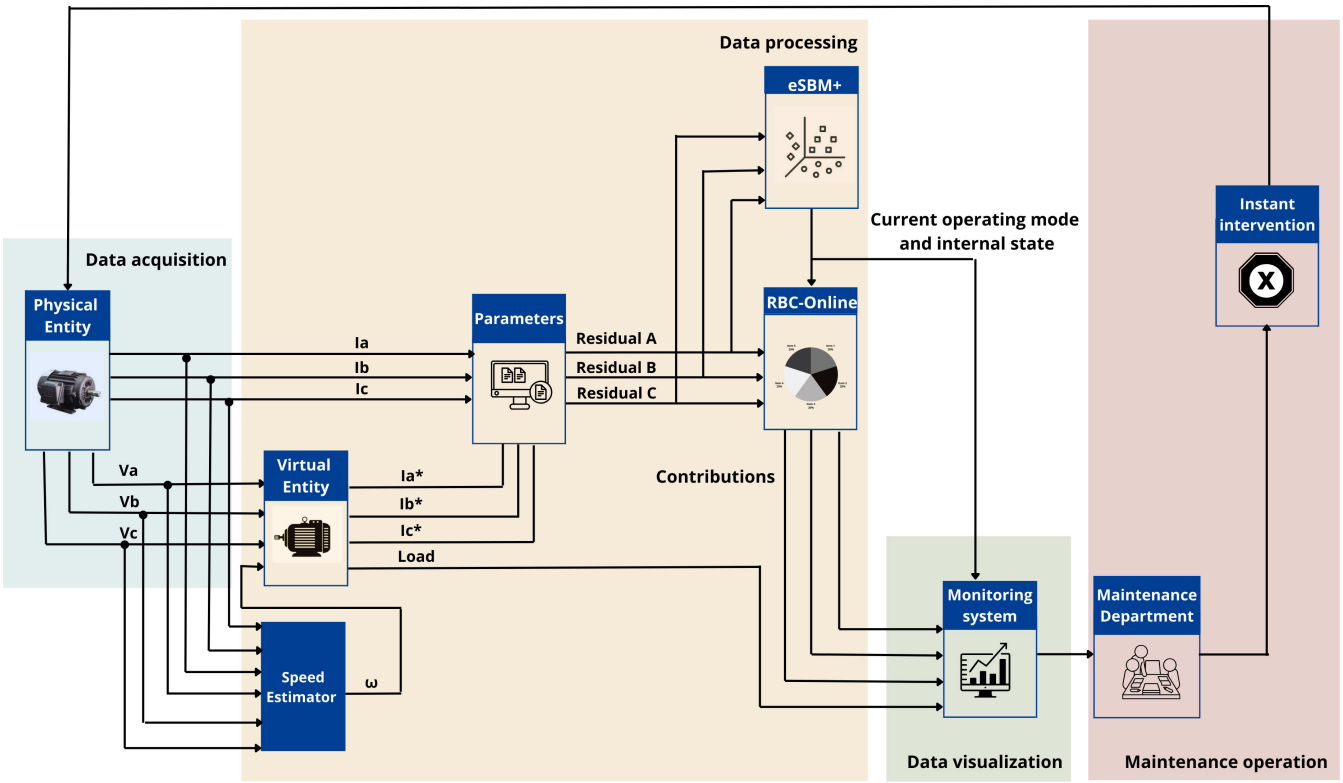


FIGURE 2. Methodology Flowchart: Expanded Representation.

The equations for the stator flux components ϕ_{qs} and ϕ_{ds} are expressed in (2) and (3).

$$\frac{d\phi_{qs}}{dt} = v_{qs} - R_s i_{qs} \tag{2}$$

$$\frac{d\phi_{ds}}{dt} = v_{ds} - R_s i_{ds} \tag{3}$$

The phase current of the machine can be determined using (4), (5) and (6).

$$\dot{I}_a = \frac{\dot{V}_a}{Z} \tag{4}$$

$$\dot{I}_b = \frac{\dot{V}_b}{Z} \tag{5}$$

$$\dot{I}_c = \frac{\dot{V}_c}{Z} \tag{6}$$

where:

$$Z = (R_s + jX_s) + \left(\frac{jX_m \cdot \left(\frac{R_r}{s} + jX_r\right)}{\frac{R_r}{s} + j(X_r + X_m)}\right) \tag{7}$$

C. MONITORED PARAMETER

Comparison between the motor current (I_a, I_b, I_c) and the value predicted by the healthy-condition model (I_a^*, I_b^*, I_c^*) involves calculating the ratio of their respective root mean square (RMS) values in (8), (9) and (10).

$$Index_a(t) = \frac{I_a}{I_a^*} \tag{8}$$

$$Index_b(t) = \frac{I_b}{I_b^*} \tag{9}$$

$$Index_c(t) = \frac{I_c}{I_c^*} \tag{10}$$

The phase residual is defined in (11), (12) and (13) as the difference between the current and historical values of this ratio, reflecting the discrepancy between the real motor behavior and the model's output.

$$Residual_a = |Index_a(t) - Index_a(t-1)| \tag{11}$$

$$Residual_b = |Index_b(t) - Index_b(t-1)| \tag{12}$$

$$Residual_c = |Index_c(t) - Index_c(t-1)| \tag{13}$$

Fig. 5 presents the three-dimensional distribution of current residuals under both healthy (H) operating conditions and initial inter-turn short-circuit (SC) across different load levels (25%, 50%, 75%, and 100%). Regardless of the applied load, two distinct clusters emerge in the 3D space: one associated with healthy operation and the other with fault. Under normal conditions (H), the residuals remain concentrated near the origin of the coordinate axes. In contrast, under faulty conditions, the monitored variables deviate significantly from this region, exhibiting a wider spread. This shift in the distribution pattern indicates that the fault introduces measurable variations in the residuals. Consequently, the residual parameter demonstrates strong potential as an effective indicator for distinguishing between

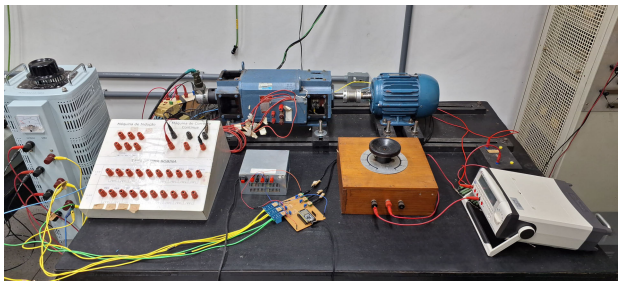


FIGURE 3. Test bed.

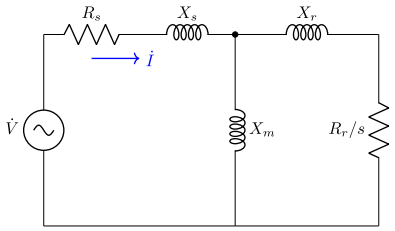


FIGURE 4. Per-phase electrical model of the induction motor.

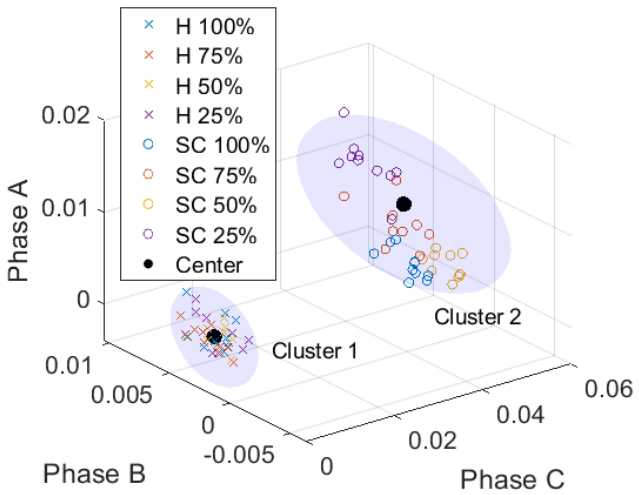


FIGURE 5. 3D Variable Trends – H vs. SC.

healthy and faulty operating states of induction motors, independent of the load level applied.

The use of a Digital Twin is essential to provide a current reference that more accurately represents the induction motor's operating state, synchronized precisely with the moment when the physical entity's current is measured. Given that the motor's equivalent circuit model depends on load conditions, the current estimated by the virtual entity adjusts correspondingly to variations in the real machine's load. Consequently, monitored variables under normal operating conditions, even across different load levels, tend to cluster into a single group (Cluster 1).

This characteristic is critical for diagnosing insulation degradation, as deviations from the healthy baseline become more pronounced and distinguishable, thereby reducing the risk of false alarms during load transitions. Regardless of the load applied to the motor, the shift from a healthy state to an initial fault condition results in the formation of a distinct cluster (Cluster 2), as illustrated in Fig. 5. It delineates the distinction between healthy operating behavior (Cluster 1) and faulty behavior across varying load conditions (Cluster 2).

In this scenario, upon fault detection, the maintenance team initiates preparations to intervene at the most appropriate time. During this interim period, if the monitoring system remains operational and load conditions fluctuate, the system continues to occupy Cluster 2, indicating fault persistence as residual behavior patterns remain consistent. Fig. 5 and Fig. 11 depict this behavior. Section III offers a comprehensive analysis of the process dynamics.

D. EVOLVING SIMILARITY-BASED MODELING PLUS (ESBM+)

The eSBM+ structure models the state of a system in a vector $x_n = [x_n(1), x_n(2), \ldots, x_n(M)]^T$ where each element represents a measurement from multiple sources. For each operational mode (i), a Memory Matrix D_i of size $L \times M$ is organized with historical data. A new state x_n is compared to previous states through a linear combination, and the estimate is given by (14).

$$\hat{x}_{(n,i)} = D_i^T w_{(n,i)} \tag{14}$$

where T is the transpose matrix of D_i, and w_n,i is the linear estimator of x_n. The estimation error is determined based on the norm defined in (15).

$$e_n = \|x_n - \hat{x}_n\|_2 \tag{15}$$

The similarity function is used to evaluate the correlation between the current state and past states, employing the Mahalanobis distance to quantify the similarity between the vectors. For improved detection accuracy, this study utilizes the Wegerich similarity function as specified in (16).

$$s_{WSF} = \frac{1}{1 + \alpha d_{ij}} \tag{16}$$

The parameter α acts as an adjustable factor that controls the decay rate of the function. During classification, the system calculates the mean relative error between the input and the estimate, then assigns the state to the group (c^*) with the smallest error. If the error falls below a defined threshold, the system associates the event with an existing group; otherwise, it classifies the event as new and captures an event window to determine whether the sample falls within the group's region. The system defines a Point Region and performs a merging process when the distance between two groups is sufficiently small, preventing redundant samples. This method updates the system's memory with new samples while maintaining clear distinctions between groups and avoiding data overload.

E. RECONSTRUCTION-BASED CONTRIBUTIONS (RBC-ONLINE)

The RBC method is used in root cause diagnostics, focusing on minimizing the effect of a variable on the fault detection index [12]. When a fault occurs in a variable x with a magnitude (f_i) along the direction (ξ_i), the estimate of the variable without the fault effects is given by (17).

$$z_i = x - \xi_i f_i \tag{17}$$

The fault detection index is determined as (18).

$$\text{Index}(z_i) = z_i^T M z_i = \|x - \xi_i f_i\|_M^2 \tag{18}$$

where M can be the Hotelling's Statistic, the combined index, or the squared prediction error. The matrix M is defined according to the index type, with the covariance matrix decomposition using PCA to obtain the principal subspace and residual matrices. After detecting the fault, the contribution of the variable x to the fault is given by (19).

$$\text{RBC}_i = \|\xi_i f_i\|_M^2 \tag{19}$$

The objective is to minimize the index (z_i) by determining the value of f_i that minimizes the equation. This is achieved by differentiating the index with respect to f_i and setting the derivative equal to zero, resulting in (20).

$$f_i = \frac{\xi_i^T M x}{\xi_i^T M \xi_i} \tag{20}$$

Substituting the value of f_i into the expression for RBC_i yields equation (21).

$$\begin{aligned} \text{RBC}_i &= \left\| \left(\xi_i^T M \xi_i\right)^{-1} \xi_i^T M x \right\|_M^2 \\ &= x^T M \xi_i \left(\xi_i^T M \xi_i\right)^{-1} \xi_i^T M x \end{aligned} \tag{21}$$

Finally, the reconstructed index is obtained by adding RBC_i to the original index, as shown in (22).

$$\text{Index}(x) = \text{Index}(z_i) + \text{RBC}_i \tag{22}$$

This final value represents the contribution of the variable x to the fault, based on the reconstruction technique.

We employed an adapted version of the RBC method for online operation, as proposed by [12]. The RBC-online approach dynamically updates the reference model by maintaining a window Q, which stores the most recent $d \in \mathbb{N}^*$ samples associated with the current group. This process occurs provided that the internal state of eSBM+ is not engaged in forming a new group.

In the event of a group transition, the reference model is first constructed using the stored window of the previous group. Subsequently, the contribution of $w \in \mathbb{N}^*$ samples is computed from the transition point onward. A new window then begins accumulating data from the newly formed group, and if another transition occurs, the procedure is repeated. Using this moving window strategy, RBC-online effectively identifies the variables that contribute most significantly to group formation and transitions, enhancing fault diagnosis and system adaptability.

F. GRAPHICAL VISUALIZATION

Fig. 6 presents the flowchart of the graphical visualization process via a smartphone or computer, enabling real-time monitoring of results. The Grafana platform is employed as the graphical visualization tool to track the operational status of the induction motor, indicating whether it is operating under normal conditions or exhibiting an inter-turn short-circuit fault.

When the cluster label exceeds 1, the system classifies the motor's condition as unhealthy and triggers a red alert. RBC-Online then quantifies the contribution of each monitored variable to the formation of the new cluster and displays the corresponding percentages in real-time. These contributions correlate directly with the phase affected by the short-circuit fault, thereby enabling precise fault localization. The Digital Twin architecture supports real-time identification of the motor's operating load level. Based on this information, the system determines whether the motor operates below, within, or beyond the limits specified by the service factor for the applied load.

A color-coded interface enhances load condition in the monitoring panel: green denotes normal operation, yellow indicates performance within the permissible service factor range, and red highlights an overload condition.

After completing the diagnostic assessment, the system initiates a new data acquisition cycle autonomously, ensuring continuous monitoring and real-time updates of operational parameters. This loop enables continuous monitoring by updating signal readings and refreshing diagnostic outputs in real-time. The platform supports local and remote access, allowing users to observe live operating data and review historical performance trends. This capability facilitates more accurate diagnostics and supports informed, timely decision-making in maintenance and operational strategies.

III. RESULT

The authors conducted a series of experiments to validate the robustness and efficiency of the proposed approach. The induction motor initially operated under fault-free conditions at 25% of rated load. After a predetermined interval, the setup introduced an inter-turn short-circuit fault. To simulate an insulation degradation scenario, we removed a tap from the winding of phase C, creating a fault equivalent to approximately 1.5% of the total number of turns in each phase. These tapped windings were externally connected in series with a resistive bank, regulating the short-circuit current at approximately 3 A, which corresponds to nearly 50% of the nominal full-load current. Under severe short-circuit conditions, the current may reach up to twice the locked-rotor current [34]. Thus, maintaining the fault-induced current below the rated level suggests insulation degradation involving only this minor part of the winding turns.

The three-dimensional dataset, constructed from current residuals generated by the induction motor's Digital Twin,

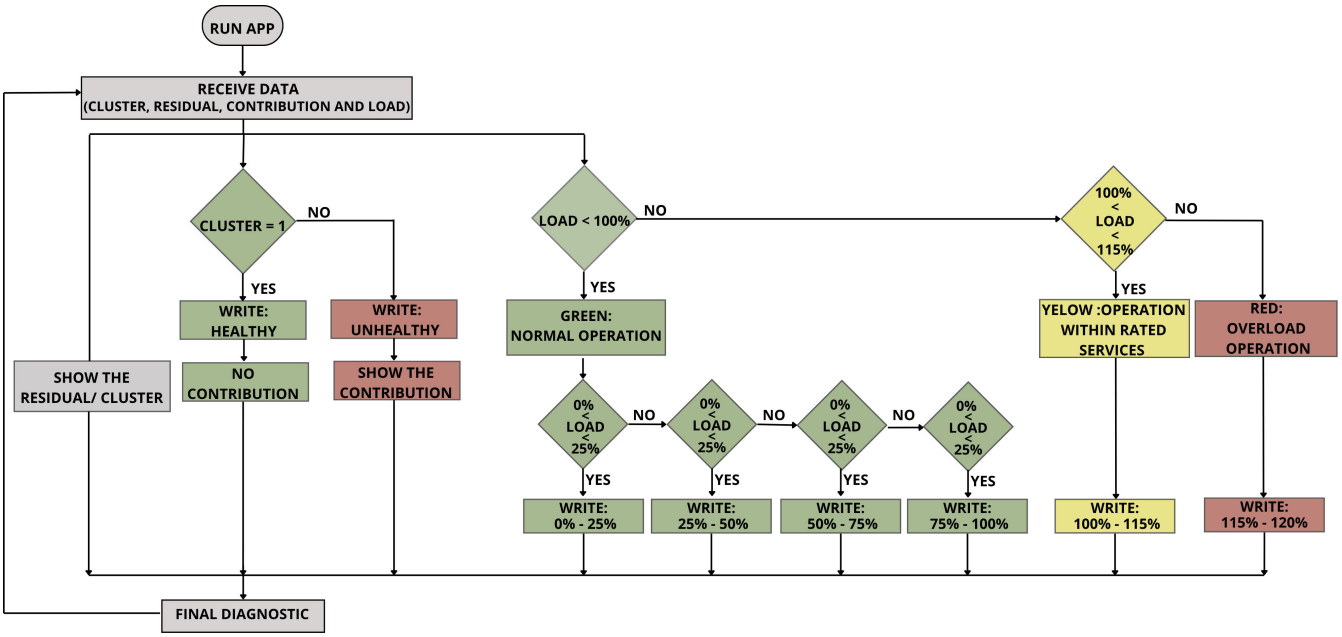


FIGURE 6. Visualization process diagram.

TABLE 2. Comparison with other methods.

Methods	Sensitivity to Low Fault Severity	Demand for Accurate Models	Fault Location Identification	Requirement for Machine Data	Computational Complexity
MCSA-ANN	Low	Low	No	Yes	Low
Flux-ANN	Low	Low	No	Yes	Low
DT-FEM	High	High	Yes	No	High
Proposed Method	**High**	**Low**	**Yes**	**No**	**Medium**

captures the temporal evolution of the monitored variables that characterize its operational state. The eSBM+ system evaluates this dataset to determine whether the machine is operating in a healthy or faulty condition. Fig. 7 presents the system functioning under normal conditions.

Fig. 8 illustrates the alert phase associated with an incipient inter-turn short-circuit fault, detected at 14:48 in an induction motor operating at 25% of its rated load. The eSBM+ algorithm identifies this fault condition through the emergence of a new cluster, signifying a deviation from the motor's baseline behavior. Diagnostic reliability improves by verifying the presence of a consistent anomaly across 12 consecutive samples before issuing an alert. This criterion effectively filters out transient noise and spurious fluctuations in the monitoring signals, ensuring that alerts reflect genuine deviations from normal operating conditions. This validation process enables the system to confirm the change in operational state within approximately two minutes. The alert, highlighted in red, signals to immediately stop the process, aiming to prevent further degradation.

RBC-Online quantifies the influence of each monitored variable on the formation of the newly detected cluster, as demonstrated in Fig. 8. This diagnostic capability facilitates accurate fault localization. In the presented case, the system determined phase C as the source of the short circuit. Consequently, the maintenance team can execute a prompt and focused response, thereby reducing downtime and preventing more severe damage to the equipment.

After the maintenance intervention to correct the short-circuit fault in the induction motor, the monitoring system demonstrates the ability to recognize the return to a healthy operating condition, as shown in Fig. 9. This behavior highlights the system's effectiveness in tracking the complete fault cycle—detection, intervention, and equipment recovery.

The proposed methodology's advantages and the critical role of the Digital Twin in diagnosing inter-turn short-circuit faults in induction motors are demonstrated by assessing how the monitoring system performs under dynamic load conditions. The motor was sequentially subjected to 25%, 50%, 75%, and 100% of its rated load at time instants 18:35:00, 18:36:23, 18:37:55, and 18:34:20, respectively, as illustrated in Fig. 10.

During each load transition, the monitored data exhibit a temporary variation until the system stabilizes under the new operating condition and resumes its prior pattern. Despite these transitional effects, the monitoring algorithm

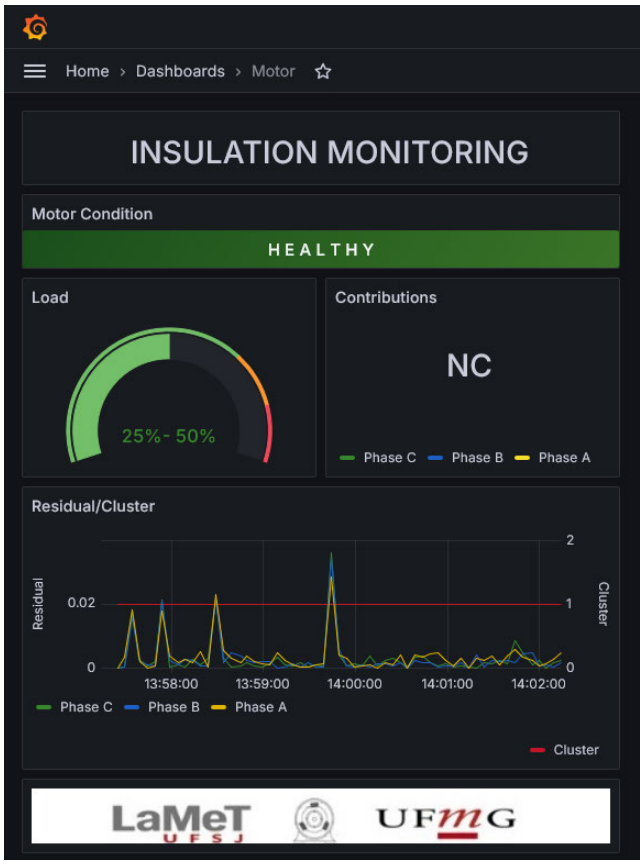


FIGURE 7. Healthy system operation.

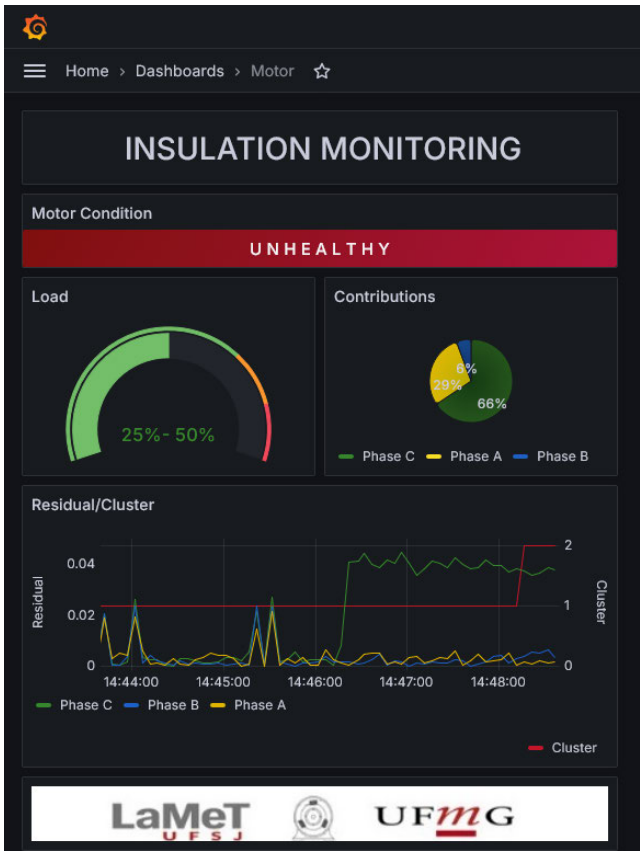


FIGURE 8. Incipient fault alert.

does not generate false alarms, confirming its resilience to operating variability. These results underscore the reliability and adaptability of the proposed approach in real-world scenarios involving load fluctuations.

Multiple scenarios served to evaluate the effectiveness of the proposed method, as detailed in Fig. 11. During the initial experimental tests, the induction motor operated under normal conditions at 50% of its rated load. Under these conditions, the system accurately classified the operational state as fault-free, grouping the data into Cluster 1. At 14:13:30, we introduced an insulation degradation between turns while maintaining the load at 50%. The system detected the fault approximately 40 seconds later (equivalent to 12 samples). From that point onward, the system allocated data to a new Cluster 2, correctly signaling the presence of the anomaly. Subsequently, at 14:14:55, we increased the load to 75% of the rated value while the degradation fault persisted. Despite this abrupt change in operating conditions, the system maintained accurate fault identification and kept the alert state active. Finally, at 14:16:00, we removed the fault, and the system resumed classification within Cluster 1, indicating the restoration of normal operating conditions. Despite abrupt transitions and incipient faults, the proposed method demonstrated high robustness by accurately identifying both the occurrence and absence of faults during load variations in online monitoring.

At 16:12:00, as illustrated in Fig. 12, the motor operated under a load that exceeded its rated power but remained within the permissible range defined by the service factor. The monitoring system detected the condition and issued a yellow-level load alert, indicating that the induction motor briefly operated within a high-load zone that remained within acceptable operational limits. This condition did not produce significant variations in the monitored variables, and the short-circuit fault diagnostic system maintained the existing cluster structure without forming new groupings. This behavior demonstrates the system's ability to distinguish between acceptable transient conditions and actual fault signatures, thereby minimizing false positives and enhancing diagnostic reliability.

Later, at 16:25:30, as shown in Fig. 13, the motor operated under a load that exceeded the limits defined by the service factor, surpassing the safe operational threshold. In response, the monitoring system identified the critical condition and issued a red-level overload alert, indicating a severe and sustained overloading condition. In contrast to the earlier event, this scenario presented a real threat to the motor's integrity, requiring immediate intervention by the maintenance team to prevent permanent damage and ensure the continuity of system operations.

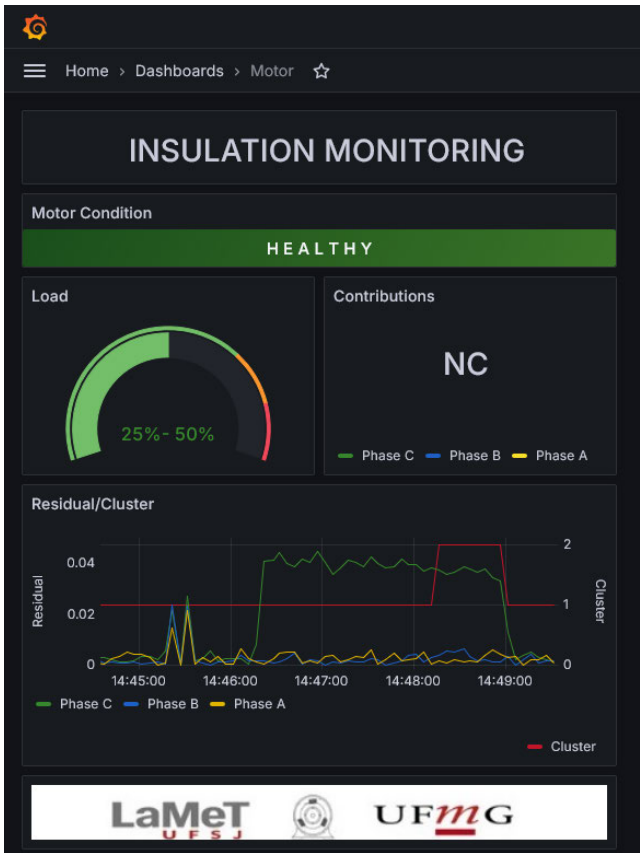


FIGURE 9. Return to healthy state.

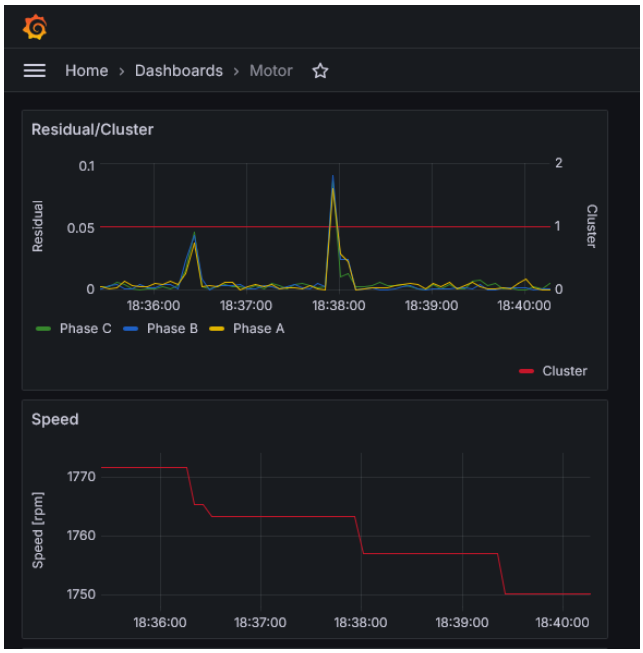


FIGURE 10. Load step changes during healthy state.

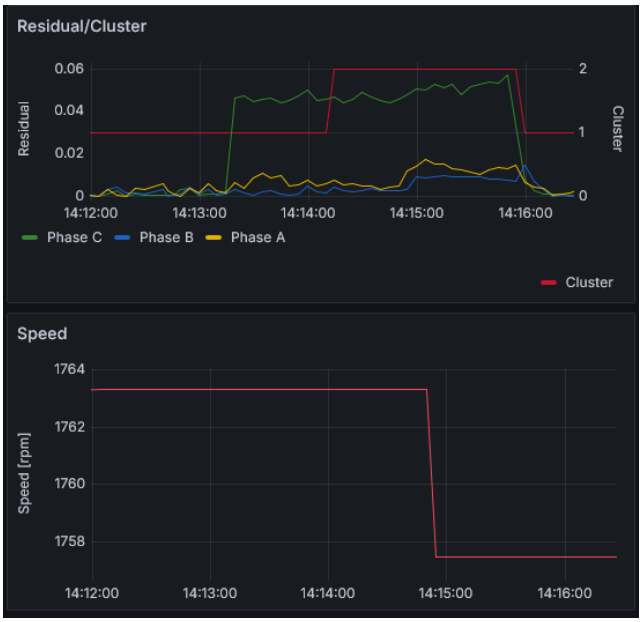


FIGURE 11. Load step changes during incipient short -circuit.

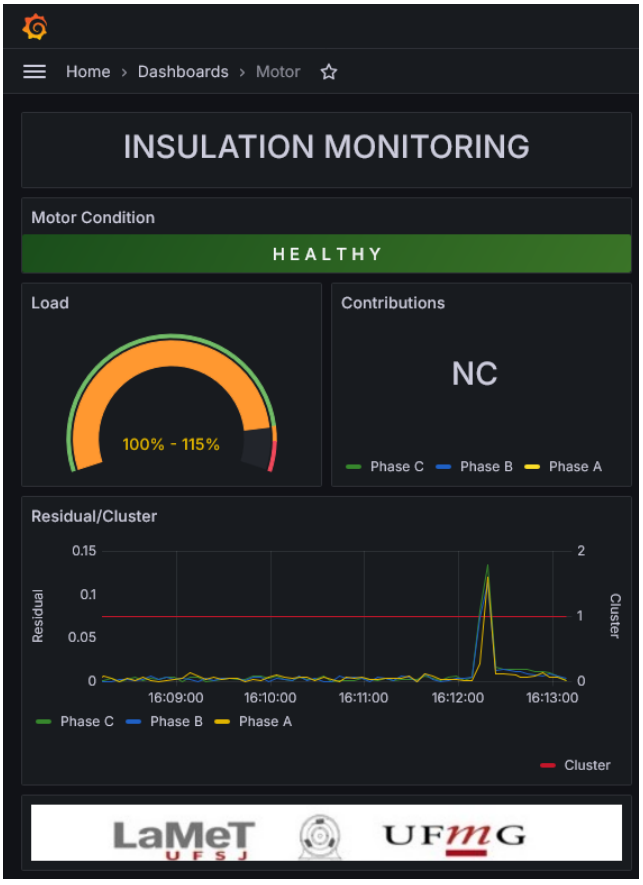


FIGURE 12. Load within service factor limits.

IV. COMPARATIVE ANALYSIS

To assess the relative strengths of the proposed method, we conducted a comparative analysis against other approaches commonly used for diagnosing inter-turn short-circuit faults, as summarized in Table 2. The evaluation considers five key criteria:

- Sensitivity to Low Fault Severity: the capability of the methodology to detect faults during early development.

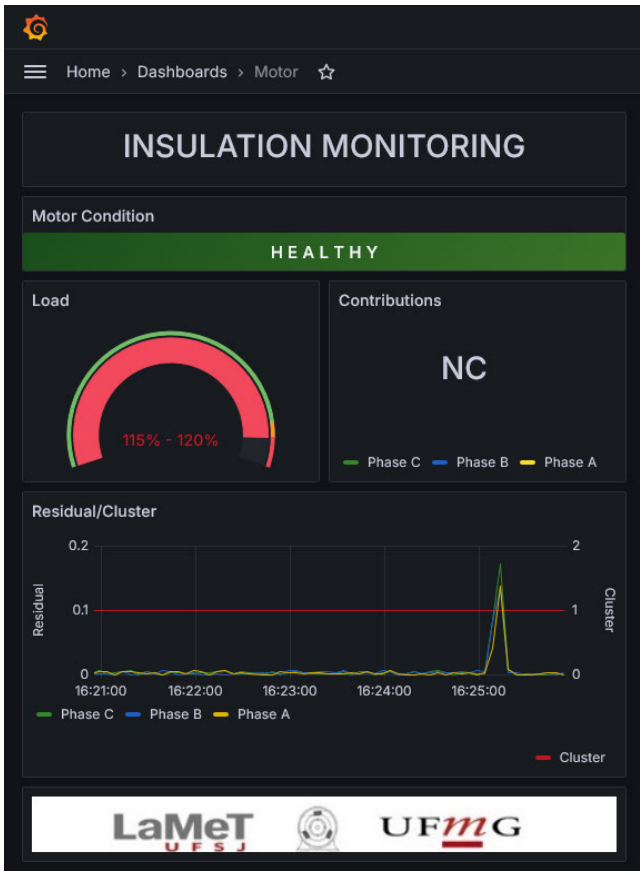


FIGURE 13. Load exceeds service factor.

Techniques that identify anomalies at the beginning of insulation degradation exhibit high sensitivity. In contrast, methods that detect faults only after the short circuit becomes well-established within a phase receive a low sensitivity rating.

- Demand for Accurate Models: the evaluation considers whether the diagnostic method requires prior machine specific knowledge, such as baseline electrical signatures or labeled data under both healthy and faulty conditions. Techniques that require detailed, precise modeling to achieve effective diagnosis receive a high demand rating. In contrast, methods that operate reliably without extensive machine modeling are classified as having low modeling requirements.
- Fault Location Identification – This binary criterion evaluates whether the method can pinpoint the exact phase in which the short circuit occurs. The technique either enables fault localization or it does not.
- Requirement for Machine-Specific Data – This criterion evaluates whether the diagnostic method relies on machine-specific prior knowledge, such as baseline electrical signatures or labeled datasets representing both faulty and healthy operating conditions. The evaluation distinguishes between methods that require such data and those that do not.
- Computational Complexity – This criterion considers the algorithmic complexity of each method. Diagnostic approaches fall along an increasing scale, beginning with signal-based techniques that involve straightforward processing, followed by model-based methods relying on steady-state representations, and culminating in finite element-based approaches, which demand significantly higher computational resources.

We evaluated the proposed method by comparing it to several established diagnostic techniques, including Motor Current Signature Analysis (MCSA), flux-based methods coupled with Artificial Neural Networks (ANN), and Digital Twin models integrated with Finite Element Methods (FEM).

The literatures [25], [35], [36], [37], [38], [39], and [40] demonstrates that MCSA detects inter-turn short circuits in three-phase induction motors by analyzing time-domain signals or specific spectral components of the stator current, known as fault signatures. This technique offers a significant advantage due to its non-intrusive nature, as it requires only stator current measurements without physical access to the motor's interior. However, when combined with intelligent algorithms like ANNs, MCSA's effectiveness depends heavily on access to extensive and representative datasets encompassing motor signals under healthy and faulty conditions across various load levels and fault severities. Gathering such comprehensive datasets from the same motor operating under diverse regimes remains a complex and critical challenge.

Flux analysis serves as a complementary diagnostic tool for three-phase induction motors, particularly for detecting inter-turn short circuits [22], [41], [42], [43], [44]. This method relies on sensors positioned to detect variations in the magnetic field or flux distribution within the stator and rotor. Although some approaches employ non-intrusive external magnetic sensors, many require physical access to the motor to install internal sensors or modify its original structure, making them intrusive. Furthermore, flux-based techniques generally involve greater implementation complexity and are more sensitive to electromagnetic interference.

Digital Twin models combined with FEM have emerged as powerful tools for generating detailed simulation databases used in detecting inter-turn short circuits in electrical machines [45], [46], [47], [48], [49], [50]. These models accurately replicate the machine's electromagnetic behavior, enabling detailed simulations under various operating conditions and fault severities. Nevertheless, developers face significant challenges related to model complexity and the high computational costs associated with these simulations.

The proposed approach employs a mathematical model based on the motor's equivalent circuit, delivering a streamlined yet effective framework for monitoring and diagnosing inter-turn short-circuit faults. Compared to FEM-based methods, it reduces implementation complexity and computational load while offering greater adaptability across different motor types. By incorporating a speed

estimator, the approach encompasses the full range of motor operating loads. This approach strikes an optimal balance among cost, performance, and practical applicability, making it highly suitable for predictive maintenance systems.

V. CONCLUSION

The Digital Twin is an essential pillar of Industry 4.0, promoting the integration of the physical and digital worlds for more efficient and strategic management of industrial assets. In this study, a virtual model facilitates the real-time diagnosis of inter-turn short-circuit faults in induction motors, thereby reducing maintenance costs and operational risks. The voltage and current sensors installed on the physical motor continuously acquire operational data, transmitting it to the computational model that simulates the machine's behavior under healthy conditions at various load levels.

The system detects discrepancies between the current predicted by the virtual model and the actual current measured in the motor. It subsequently utilizes the evolving system eSBM+ to evaluate the motor's operational state and identify faults. The RBC-Online technique is then applied to pinpoint the specific phase affected by the short circuit.

This modeling framework effectively captures the impacts of load variations, ensuring robust and reliable fault detection. Its ability to handle non-ideal operating conditions further enhances its relevance to industrial environments. The experimental results demonstrate that this approach enables early fault detection, promotes data-driven maintenance strategies, and extends the motor's lifespan. Its ease of implementation supports the broader adoption of online monitoring systems and provides a foundation for detecting different motor faults in the future. Notably, the proposed system is capable of detecting low insulation levels using low-cost hardware without compromising data reliability—highlighting its potential for accessible and scalable diagnostic solutions.

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REFERENCES

[1] Y. Lu, C. Liu, K. I.-K. Wang, H. Huang, and X. Xu, "Digital twin-driven smart manufacturing: Connotation, reference model, applications and research issues," *Robot. Comput.-Integr. Manuf.*, vol. 61, Feb. 2020, Art. no. 101837.

[2] D. M. Botín-Sanabria, A.-S. Mihaita, R. E. Peimbert-García, M. A. Ramírez-Moreno, R. A. Ramírez-Mendoza, and J. D. J. Lozoya-Santos, "Digital twin technology challenges and applications: A comprehensive review," *Remote Sens.*, vol. 14, no. 6, p. 1335, Mar. 2022.

[3] A. Sharma, E. Kosasih, J. Zhang, A. Brintrup, and A. Calinescu, "Digital twins: State of the art theory and practice, challenges, and open research questions," *J. Ind. Inf. Integr.*, vol. 30, Nov. 2022, Art. no. 100383.

[4] Q. Qi and F. Tao, "Digital twin and big data towards smart manufacturing and Industry 4.0: 360 degree comparison," *IEEE Access*, vol. 6, pp. 3585–3593, 2018.

[5] M. Liu, S. Fang, H. Dong, and C. Xu, "Review of digital twin about concepts, technologies, and industrial applications," *J. Manuf. Syst.*, vol. 58, pp. 346–361, Jan. 2021.

[6] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.

[7] I. Errandonea, S. Beltrán, and S. Arrizabalaga, "Digital twin for maintenance: A literature review," *Comput. Ind.*, vol. 123, Dec. 2020, Art. no. 103316.

[8] L. Zhao, T. Xie, Y. Wei, Y. Liu, and Y. Qin, "Overview of digital twin-driven rotating machinery fault diagnosis: Status and trends," *Meas. Sci. Technol.*, vol. 36, no. 5, May 2025, Art. no. 052001.

[9] M. Xia, H. Shao, D. Williams, S. Lu, L. Shu, and C. W. de Silva, "Intelligent fault diagnosis of machinery using digital twin-assisted deep transfer learning," *Rel. Eng. Syst. Saf.*, vol. 215, Nov. 2021, Art. no. 107938.

[10] Z. Chen, D. Liang, S. Jia, L. Yang, and S. Yang, "Incipient interturn short-circuit fault diagnosis of permanent magnet synchronous motors based on the data-driven digital twin model," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 11, no. 3, pp. 3514–3524, Jun. 2023.

[11] M. Soori, F. K. G. Jough, R. Dastres, and B. Arezoo, "AI-based decision support systems in Industry 4.0, a review," *J. Economy Technol.*, 2024, doi: 10.1016/j.ject.2024.08.005. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2949948824000374

[12] N. M. Almeida and W. M. Caminhas, "Proposta de Framework com Foco em métodos evolutivos para detecçao, diagnóstico e causa Raiz de Falhas em processos industriais," in *Proc. Anais do Congresso Brasileiro de Automática*, pp. 1–6, Dec. 2020.

[13] N.-C. Yang and J.-M. Yang, "Fault classification in distribution systems using deep learning with data preprocessing methods based on fast dynamic time warping and short-time Fourier transform," *IEEE Access*, vol. 11, pp. 63612–63622, 2023.

[14] A. Abdulkareem, T. Anyim, O. Popoola, J. Abubakar, and A. Ayoade, "Prediction of induction motor faults using machine learning," *Heliyon*, vol. 11, no. 1, Jan. 2025, Art. no. e41493.

[15] V. G. D. Araújo, A. O.-S. Bissiriou, J. M. M. Villanueva, E. R. L. Villarreal, A. O. Salazar, R. D. A. Teixeira, and D. A. D. M. Fonsêca, "Monitoring and diagnosing faults in induction motors' three-phase systems using NARX neural network," *Energies*, vol. 17, no. 18, p. 4609, Sep. 2024.

[16] H. Bechara, R. Ibrahim, R. Zemouri, B. Kedjar, A. Merkhouf, A. Tahan, and K. Al-Haddad, "Review of artificial intelligence methods for faults monitoring, diagnosis, and prognosis in hydroelectric synchronous generators," *IEEE Access*, vol. 12, pp. 173599–173617, 2024.

[17] G. Xu, M. Liu, Z. Jiang, W. Shen, and C. Huang, "Online fault diagnosis method based on transfer convolutional neural networks," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 2, pp. 509–520, Feb. 2020.

[18] H. Al-Sahaf, Y. Bi, Q. Chen, A. Lensen, Y. Mei, Y. Sun, B. Tran, B. Xue, and M. Zhang, "A survey on evolutionary machine learning," *J. Roy. Soc. New Zealand*, vol. 49, no. 2, pp. 205–228, 2019.

[19] T. Amarbayasgalan and K. H. Ryu, "Unsupervised feature-construction-based motor fault diagnosis," *Sensors*, vol. 24, no. 10, p. 2978, May 2024.

[20] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, Apr. 2019.

[21] A. Angelopoulos, E. T. Michailidis, N. Nomikos, P. Trakadas, A. Hatziefremidis, S. Voliotis, and T. Zahariadis, "Tackling faults in the Industry 4.0 era—A survey of machine-learning solutions and key aspects," *Sensors*, vol. 20, no. 1, p. 109, 2019.

[22] P. C. M. L. Filho, L. M. R. Baccarini, F. B. Batista, and A. C. Araújo, "Orbit analysis from a stray flux full spectrum for induction machine fault detection," *IEEE Sensors J.*, vol. 21, no. 14, pp. 16152–16161, Jul. 2021.

[23] W. F. Godoy, I. N. da Silva, A. Goedtel, and R. H. C. Palácios, "Evaluation of stator winding faults severity in inverter-fed induction motors," *Appl. Soft Comput.*, vol. 32, pp. 420–431, Jul. 2015.

[24] V. Fernandez-Cavero, D. Morinigo-Sotelo, O. Duque-Perez, and J. Pons-Llinares, "Fault detection in inverter-fed induction motors in transient regime: State of the art," in *Proc. IEEE 10th Int. Symp. Diag. Electr. Mach., Power Electron. Drives (SDEMPED)*, Sep. 2015, pp. 205–211.

[25] P. Gangsar and R. Tiwari, "Signal based condition monitoring techniques for fault detection and diagnosis of induction motors: A state-of-the-art review," *Mech. Syst. Signal Process.*, vol. 144, Oct. 2020, Art. no. 106908.

[26] M. Eftekhari, M. Moallem, S. Sadri, and M.-F. Hsieh, "Online detection of induction motor's stator winding short-circuit faults," *IEEE Syst. J.*, vol. 8, no. 4, pp. 1272–1282, Dec. 2014.

[27] S. Grubic, J. M. Aller, B. Lu, and T. G. Habetler, "A survey on testing and monitoring methods for stator insulation systems of low-voltage induction machines focusing on turn insulation problems," *IEEE Trans. Ind. Electron.*, vol. 55, no. 12, pp. 4127–4136, Dec. 2008.

[28] A. Siddique, G. S. Yadava, and B. Singh, "A review of stator fault monitoring techniques of induction motors," *IEEE Trans. Energy Convers.*, vol. 20, no. 1, pp. 106–114, Mar. 2005.
[29] O. Alshorman and A. Alshorman, "A review of intelligent methods for condition monitoring and fault diagnosis of stator and rotor faults of induction machines," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 11, no. 4, p. 2820, Aug. 2021.
[30] J. Lee, B. Bagheri, and H.-A. Kao, "A cyber-physical systems architecture for industry 4.0-based manufacturing systems," *Manuf. Lett.*, vol. 3, pp. 18–23, Jan. 2015.
[31] M. B. Yassein, M. Q. Shatnawi, S. Aljwarneh, and R. Al-Hatmi, "Internet of Things: Survey and open issues of MQTT protocol," in *Proc. Int. Conf. Eng. MIS (ICEMIS)*, May 2017, pp. 1–6.
[32] Y. Hao, X. Qin, Y. Chen, Y. Li, X. Sun, Y. Tao, X. Zhang, and X. Du, "TS-benchmark: A benchmark for time series databases," in *Proc. IEEE 37th Int. Conf. Data Eng. (ICDE)*, Apr. 2021, pp. 588–599.
[33] Espressif Systems. (2024). *ESP-IDF Programming Guide: ADC Calibration*. Accessed: Jul. 10, 2025. [Online]. Available: https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-reference/peripherals/adc_calibration.html
[34] R. M. Tallam, S. B. Lee, G. C. Stone, G. B. Kliman, J. Yoo, T. G. Habetler, and R. G. Harley, "A survey of methods for detection of stator-related faults in induction machines," *IEEE Trans. Ind. Appl.*, vol. 43, no. 4, pp. 920–933, Jul./Aug. 2007.
[35] L. A. Pereira, D. da Silva Gazzana, and L. F. A. Pereira, "Motor current signature analysis and fuzzy logic applied to the diagnosis of short-circuit faults in induction motors," in *Proc. 31st Annu. Conf. IEEE Ind. Electron. Soc. (IECON)*, Nov. 2005, p. 6.
[36] A. Mejia-Barron, G. Tapia-Tinoco, J. R. Razo-Hernandez, M. Valtierra-Rodriguez, and D. Granados-Lieberman, "A neural network-based model for MCSA of inter-turn short-circuit faults in induction motors and its power hardware in the loop simulation," *Comput. Electr. Eng.*, vol. 93, Jul. 2021, Art. no. 107234.
[37] G. Niu, X. Dong, and Y. Chen, "Motor fault diagnostics (don't short) based on current signatures: A review," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–19, 2023.
[38] A. H. Bonnett and G. C. Soukup, "Cause and analysis of stator and rotor failures in three-phase squirrel-cage induction motors," *IEEE Trans. Ind. Appl.*, vol. 28, no. 4, pp. 921–937, Jul./Aug. 1992.
[39] A. Allal and A. Khechekhouche, "Diagnosis of induction motor faults using the motor current normalized residual harmonic analysis method," *Int. J. Electr. Power Energy Syst.*, vol. 141, Oct. 2022, Art. no. 108219.
[40] G. M. Joksimovic and J. Penman, "The detection of inter-turn short circuits in the stator windings of operating motors," *IEEE Trans. Ind. Electron.*, vol. 47, no. 5, pp. 1078–1084, Oct. 2000.
[41] P. C. M. L. Filho, D. C. Santos, F. B. Batista, and L. M. R. Baccarini, "Axial stray flux sensor proposal for three-phase induction motor fault monitoring by means of orbital analysis," *IEEE Sensors J.*, vol. 20, no. 20, pp. 12317–12325, Oct. 2020.
[42] F. B. Batista, P. C. M. L. Filho, R. Pederiva, and V. A. D. Silva, "An empirical demodulation for electrical fault detection in induction motors," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 3, pp. 559–569, Mar. 2016.
[43] A. Yazidi, D. Thailly, H. Henao, R. Romary, G.-A. Capolino, and J.-F. Brudny, "Detection of stator short-circuit in induction machines using an external leakage flux sensor," in *Proc. IEEE Int. Conf. Ind. Technol. (IEEE ICIT)*, vol. 1, Dec. 2004, pp. 166–169.
[44] M. Schroedl and R. S. Wieser, "EMF-based rotor flux detection in induction motors using virtual short circuits," *IEEE Trans. Ind. Appl.*, vol. 34, no. 1, pp. 142–147, Jan./Feb.1998.
[45] S. Bouzid, P. Viarouge, and J. Cros, "Real-time digital twin of a wound rotor induction machine based on finite element method," *Energies*, vol. 13, no. 20, p. 5413, Oct. 2020.
[46] R. R. Singh, G. Bhatti, D. Kalel, I. Vairavasundaram, and F. Alsaif, "Building a digital twin powered intelligent predictive maintenance system for industrial AC machines," *Machines*, vol. 11, no. 8, p. 796, Aug. 2023.
[47] C. Rojas, M. G. Melero, M. F. Cabanas, J. M. Cano, G. A. Orcajo, and F. Pedrayes, "Finite element model for the study of inter-turn short circuits in induction motors," in *Proc. IEEE Int. Symp. Diag. Electr. Mach., Power Electron. Drives*, Sep. 2007, pp. 415–419.
[48] N. P. Kumar and T. B. Isha, "FEM based electromagnetic signature analysis of winding inter-turn short-circuit fault in inverter fed induction motor," *CES Trans. Electr. Mach. Syst.*, vol. 3, no. 3, pp. 309–315, Sep. 2019.
[49] Y. Bensalem and M. N. Abdelkrim, "Modeling and simulation of induction motor based on finite element analysis," *Int. J. Power Electron. Drive Syst. (IJPEDS)*, vol. 7, no. 4, p. 1100, Dec. 2016.
[50] D. Díaz, M. C. Amaya, and A. Paz, "Inter-turn short-circuit analysis in an induction machine by finite elements method and field tests," in *Proc. 20th Int. Conf. Electr. Mach.*, Sep. 2012, pp. 1757–1763.



ANA C. S. ARAÚJO received the B.Sc. and M.Sc. degrees in electrical engineering from the Federal University of São João del Rei (UFSJ), Brazil, in 2020 and 2021, respectively. She is currently pursuing the Ph.D. degree in electrical engineering with the Federal University of Minas Gerais (UFMG). Her research interests include diagnosis and fault detection, signal processing, predictive maintenance, electric drives, and machine learning.



LANE M. RABELO (Member, IEEE) received the B.Sc. degree (Hons.) in electrical engineering from Faculdade de Engenharia Elétrica de São João del Rei, [recognized today as Federal University of São João del Rei (UFSJ)], the M.Sc. degree in electrical engineering from the Federal University of Itajubá (UNIFEI), Itajubá, Brazil, and the Ph.D. degree in electrical engineering from the Federal University of Minas Gerais (UFMG). She is currently a Professor of electrical engineering with UFSJ. She has been teaching electrical machines, since 1990. Her main research interests include machine learning techniques and drive control and diagnosis in electrical machines.



PAULO C. M. LAMIM FILHO received the degree in electrical engineering from the Federal University of São João del Rei (UFSJ), São João del Rei, Brazil, in 2000, and the M.Sc. and Ph.D. degrees from the State University of Campinas (UNICAMP), Brazil, in 2003 and 2007, respectively. Since 2010, he has been a Professor with the Electrical Engineering Course, UFSJ. His research interests include online condition monitoring and the diagnosis of electrical rotating machines.



WALMIR M. CAMINHAS received the B.Sc. and M.Sc. degrees in electrical engineering from the Federal University of Minas Gerais (UFMG), Belo Horizonte, Brazil, in 1987 and 1989, respectively, and the Ph.D. degree in electrical engineering from the University of Campinas, Campinas, Brazil, in 1997. Since 1989, he has been a Professor with the Department of Electronics Engineering, UFMG. His research interests include computational intelligence, fault detection and diagnosis, and nonlinear systems modeling.

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - ROR identifier: 00x0ma614